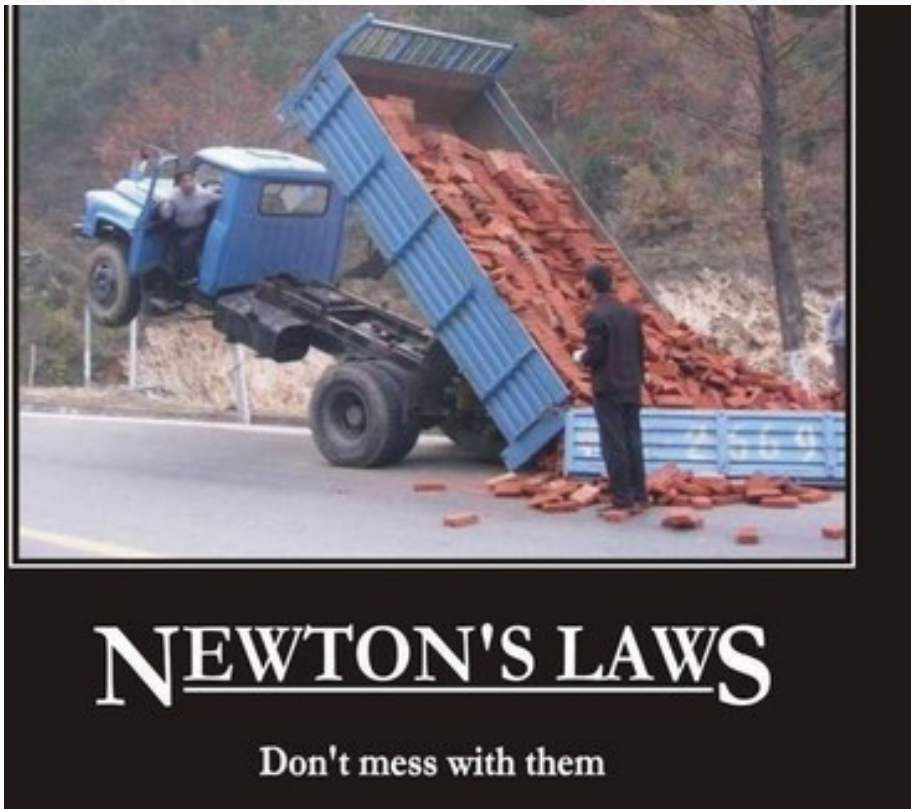


Welcome Physics 2053 Fall 25



Physics without Calculus - PHYS 2053 Fall 25

- Slides, tutorials for HW, solutions for HW can be found in my dropbox (see link in canvas – Let's visit canvas- print out equation sheet)
- Syllabus, assignments, link to resources can be found in canvas.
- I record the lectures and upload them to my Utube Channel. I still highly recommend to attend in person. For class discussion.

Relative weight will be assigned as follow:

Homework 18 % (Tutorials in dropbox. Prep for test. Don't cheat)

Tests 60% (test1, test2, final are in person. On paper. 20% each)

pop-up quizzes 12% (pop quiz at the end of each class. Open notes. Discussion with peers encouraged. Based on previous lecture)

take home test 10% (test3. Online. Window of 2 hours. Open notes)

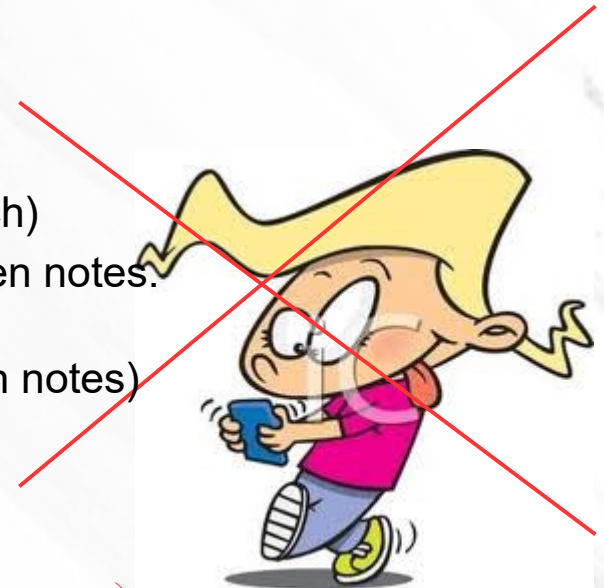
Attendance 5% → **5 points extra!**

Contact: vlankar@fiu.edu

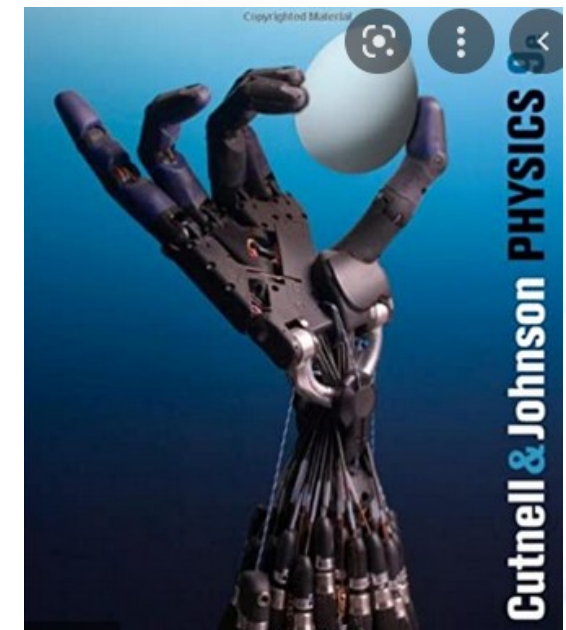
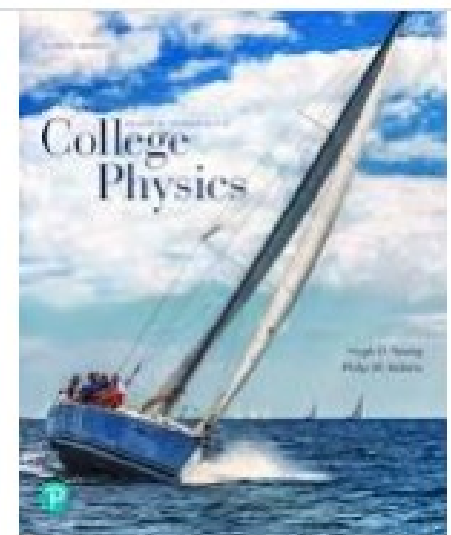
Lectures: Tu/Th 2:00PM-3:40PM @ Lib265

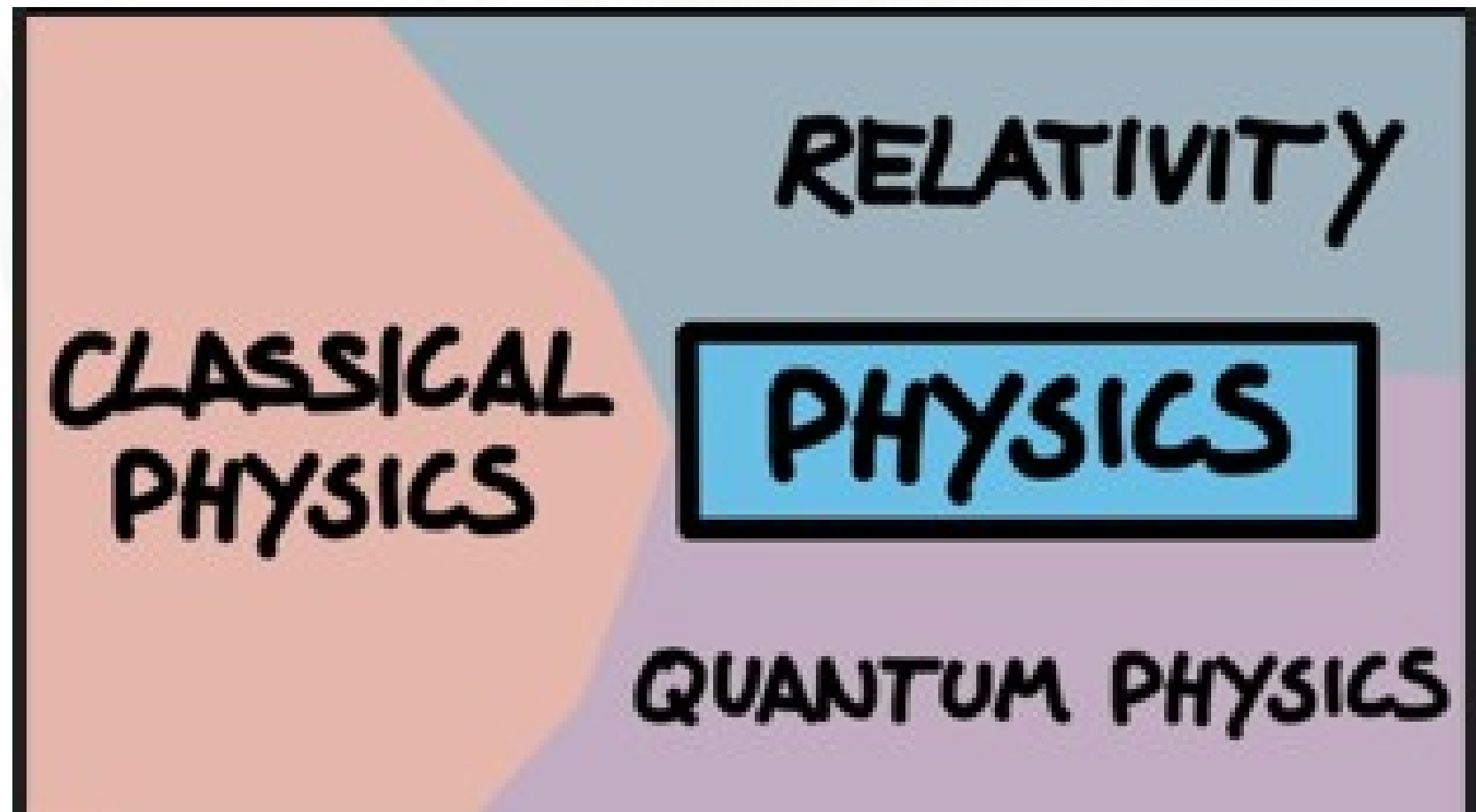
Office hours: after class and M/W 11:15AM sharp MS150

Bring a TI83/84 or at least a TI 30 in class



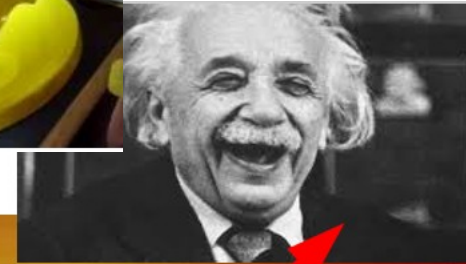
LECTURES	CHAPTERS
Unit 0: Math review Proportions, scientific notation	
Unit 1: Kinematics 1D, Free-fall kinematics equations, graphical analysis, free-fall	Chapter 2 2.1 → 2.6
Unit 2: Kinematics 2D - Components of forces, Projectile motion, circular motion, rotational kinematics	Chapter 2 2.1 → 2.6 Chapter 1 1.7, 1.8
Unit 3: Forces, Vectors Adding forces algebraically, types of forces (weight, normal, tension and frictional forces), static equilibrium, rotational equilibrium (torques), center of mass	Chapter 4 4.1, 4.6 Chapter 5 5.1 → 5.3 Chapter 6 6.4 Chapter 8 8.6 Chapter 10 10.1, 10.6
Unit 4 : Newton's laws - impulse Newton's laws, impulse-momentum theorem, Newton's second law in 2D, gravitational law, weightlessness	Chapter 4 4.2 → 4.5 Chapter 6 6.3, 6.5 Chapter 8 8.5
Unit 5: conservation of momentum Conservation of linear momentum, collision	Chapter 8 8.1 → 8.5
Unit 6: Energy, conservation Work and energy, KE, PE, conservation of energy, conservative and non-conservative forces. Power	Chapter 7 7.1 → 7.8
Unit 7: rotational kinematics - rotational dynamics, moment of inertia, angular kinetic energy	Chapter 9 9.1 → 9.5
Unit 8: rotational dynamics - angular momentum Newton's second law, angular momentum, conservation	Chapter 10 10.2 → 10.5
Unit 9: Fluids See playlist in utube channel Density, pressure, Pascal's law, buoyancy. Equation of continuity, Bernoulli	Chapter 13 13.1, 13.2, 13.3 13.5 → 13.7
Unit 10: SHM - Waves See playlist in utube channel Simple harmonic motion. Elastic potential energy. Elastic deformation. Wave properties. Sound	Chapter 11 11.1 → 11.3 Chapter 12 12.1, 12.2, 12.3 12.9, 12.10



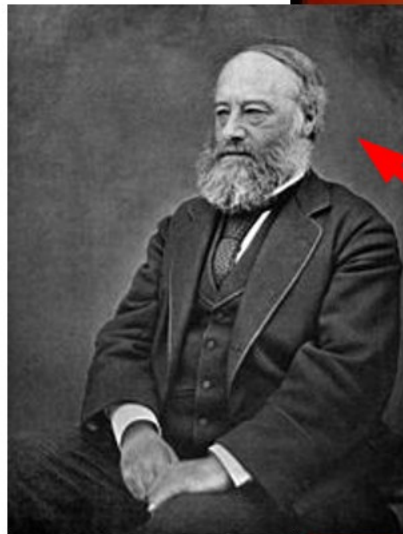
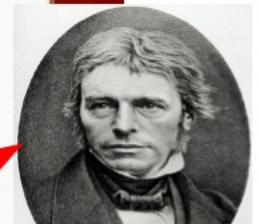




Demo: free-fall
EM /



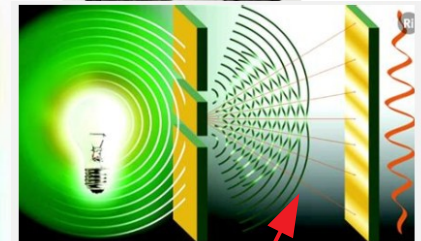
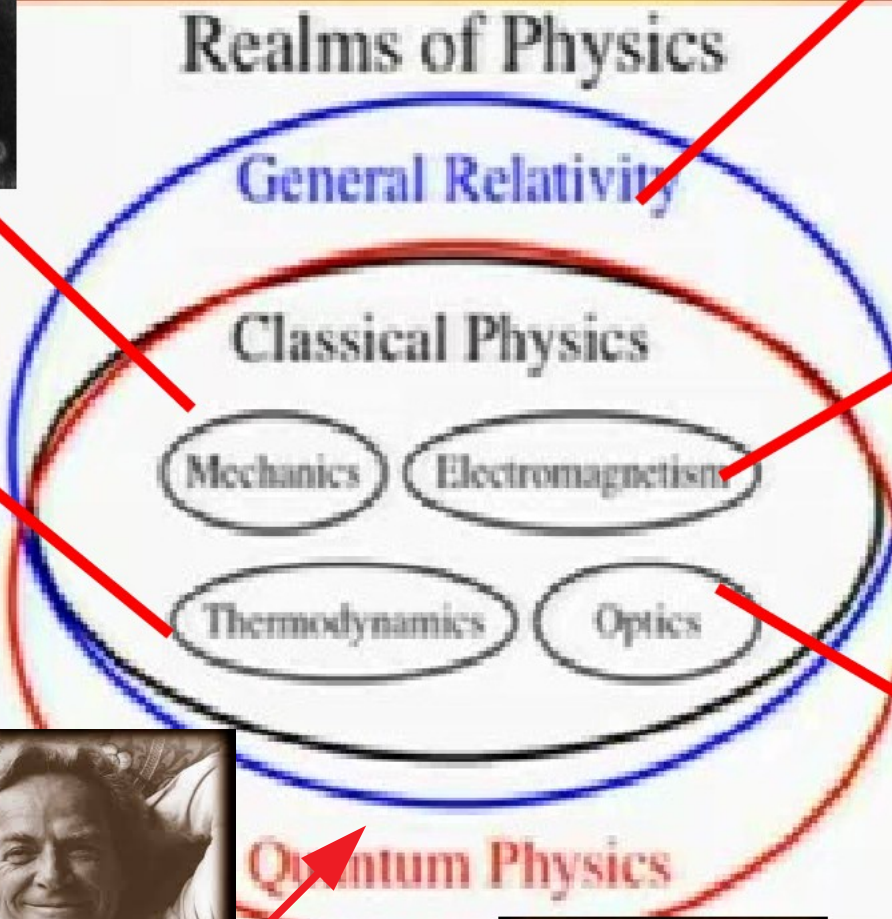
Michael Faraday



James Prescott
Joule



Richard Feynman



Thomas Young

<https://youtu.be/Iuv6hY6zsd0?t=281>

Physics experiments involve the measurement of a variety of physical quantities.

These measurements should be accurate and reproducible.

The first step in ensuring accuracy and reproducible is defining the **units** in which the measurements are made.

Units and derived units

SI Base Units

Base Quantity	Name	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Important:

Speed is in m/s
and m/s is noted m s^{-1}

Acceleration is in m/s/s
And is also noted m s^{-2}

Distance (displacement) is in m

SI Derived Units

Derived Quantity	Name	Symbol	Equivalent SI units
Frequency	hertz	Hz	s^{-1}
Force	newton	N	$\text{m} \cdot \text{kg} \cdot \text{s}^{-2}$
Pressure	pascal	Pa	N/m^2
Energy	joule	J	$\text{N} \cdot \text{m}$
Power	watt	W	J/s
Electric charge	coulomb	C	$\text{s} \cdot \text{A}$
Electric potential	volt	V	W/A
Electric resistance	ohm	Ω	V/A
Celsius temperature	degree Celsius	$^{\circ}\text{C}$	K°

Important:

Units can give away equations.

Force is $\text{kg m/s/s} \rightarrow$

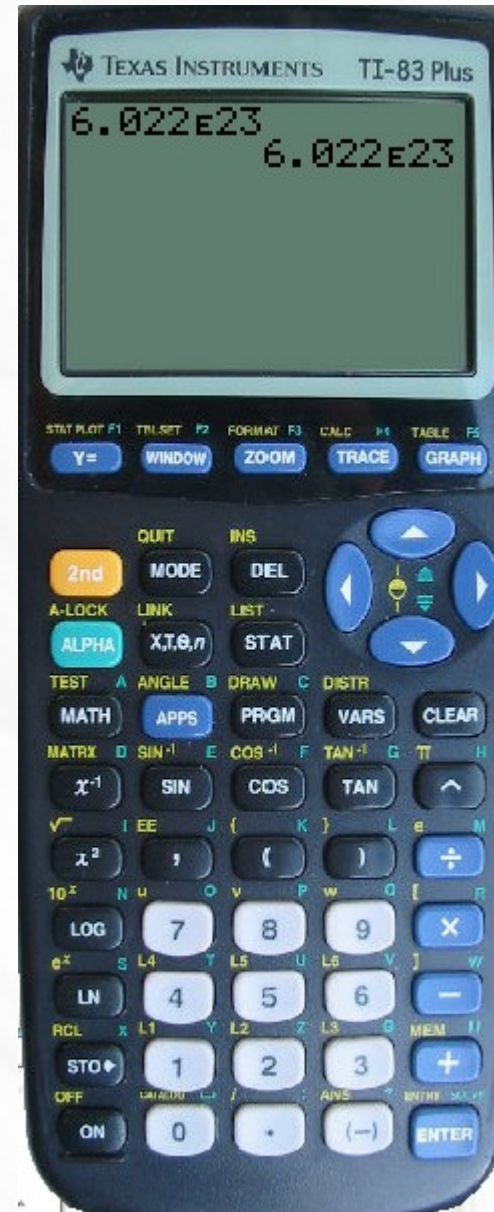
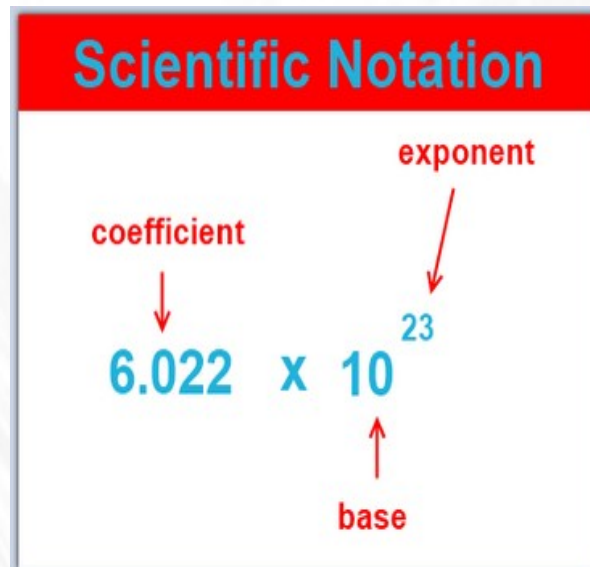
Find the equation for force

find the equation for power

find the equation for pressure

Find the equation for energy

Scientific notation → [assignment_sci_not.pdf](#)



Following is a list of prefixes and their meanings that are often used in conjunction with SI units:

Multiple	Prefix	Symbol
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n

Handwritten examples and conversions:

- $\text{kilo (k)} = 10^3$
 - $1 \text{ km} = 10^3 \text{ m}$
 - $1 \text{ kg} = 10^3 \text{ g}$
- $\text{milli (m)} = 10^{-3}$
 - $5 \text{ ml} = 5 \times 10^{-3} \text{ l}$
 - $5 \text{ mm} = 5 \times 10^{-3} \text{ m}$
 - $5 \text{ mg} = 5 \times 10^{-3} \text{ g}$
- $\text{nano (n)} = 10^{-9}$
 - $7 \text{ ns} = 7 \times 10^{-9} \text{ sec}$
 - $7 \text{ nm} = 7 \times 10^{-9} \text{ m}$
- $\text{tera (T)} = 10^{12}$
 - $5 \text{ TJ} = 5 \times 10^{12} \text{ joules}$
- $\text{mega (M)} = 10^6$
 - $25 \text{ MB} = 25 \times 10^6 \text{ B}$
- $\text{centi (c)} = 10^{-2}$
 - $5 \text{ cm} = 5 \times 10^{-2} \text{ m}$
 - $5 \text{ cg} = 5 \times 10^{-2} \text{ g}$
- $\text{micro (}\mu\text{)} = 10^{-6}$
 - $3 \mu\text{m} = 3 \times 10^{-6} \text{ m}$
 - $3 \mu\text{g} = 3 \times 10^{-6} \text{ g}$

Examples: In the news 8/13/23 91 euros per MWh (france)

Unit for electric energy - (\$160 in Florida)

Protons in the LHC get about 6.5 tera-electronvolts or TeV.



To print out and to keep.

$$1\text{km} = 1,000\text{m} (10^3 \text{ meters})$$

$$1\text{m} = 100\text{cm} (10^2 \text{ centimeters})$$

$$1\text{cm} = 10\text{mm} (10 \text{ millimeters})$$

$$1\text{km} = 100,000 \text{ cm} (10^5 \text{ cm})$$

$$1\text{mm} = 1000 \mu\text{m} (10^3 \text{ micrometers})$$

$$1\text{m} = 1000 \text{ mm} (10^3 \text{ mm})$$

$$1\text{m} = 1,000,000 \mu\text{m} (10^6 \mu\text{m})$$

note that: if $1\text{m} = 10^6 \mu\text{m}$ then $1\mu\text{m} = 10^{-6} \text{ m}$

$$1\text{mm} = 10^{-3} \text{ m} \quad 1\text{cm} = 10^{-2} \text{ m}$$

$$1 \text{ mile} = 1,609\text{m}$$

$$1 \text{ mile} = 1.6 \text{ km}$$

$$10 \text{ N} = 2.2 \text{ pounds}$$

$$1\text{min} = 60\text{s}$$

$$1\text{hour} = 60\text{min} = 3600\text{s}$$

$$1 \text{ day} = 24 \text{ h} = 24 \times 3600 \text{ s} = 86,400\text{s}$$

$$1\text{kg} = 1000 \text{ g}$$

$$1\text{g} = 1,000\text{mg}$$

Following is a list of prefixes and their meanings that are often used in conjunction with SI units:

Multiple	Prefix	Symbol
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10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n

Worth remembering:

$$1\text{kg} \rightarrow 2.2 \text{ pounds}$$

$$1\text{m/s} \rightarrow 2 \text{ mph (or } 3.6 \text{ km/h)}$$

$$3.1 \text{ miles is } 5\text{K}$$

$$6.2 \text{ miles is } 10\text{K}$$

(runkeeper)

The Universe by Orders of Magnitude

Size or Distance	(m)	Mass	(kg)	Time Interval	(s)
Proton	10^{-15}	Electron	10^{-30}	Time for light to cross nucleus	10^{-23}
Atom	10^{-10}	Proton	10^{-27}	Period of visible light radiation	10^{-15}
Virus	10^{-7}	Amino acid	10^{-25}	Period of microwaves	10^{-10}
Giant amoeba	10^{-4}	Hemoglobin	10^{-22}	Half-life of muon	10^{-6}
Walnut	10^{-2}	Flu virus	10^{-19}	Period of highest audible sound	10^{-4}
Human	10^0	Giant amoeba	10^{-8}	Period of human heartbeat	10^0
Highest mountain	10^4	Raindrop	10^{-6}	Half-life of free neutron	10^3
Earth	10^7	Ant	10^{-2}	Period of earth's rotation	10^5
Sun	10^9	Human	10^2	Period of earth's revolution around sun	10^7
Distance from earth to sun	10^{11}	Saturn V rocket	10^6	Lifetime of human	10^9
Solar system	10^{13}	Pyramid	10^{10}	Half-life of plutonium-239	10^{12}
Distance to nearest star	10^{16}	Earth	10^{24}	Lifetime of mountain range	10^{15}
Milky Way galaxy	10^{21}	Sun	10^{30}	Age of earth	10^{17}
Visible universe	10^{26}	Milky Way galaxy	10^{41}	Age of universe	10^{18}
		Universe	10^{52}		